

CMS TEAM PROJECT · CHAIR OF MULTIMEDIA TECHNOLOGY · TU DRESDEN

Choosing the figures a table can honestly carry

A chart-selection agent for arbitrary uploaded datasets: what we built, the three claims we had to walk back, and five experiments — with the speaker script for presenting them.

Branch feat/chart-selection · 22 commits ahead of main · 50 files, +13,623 lines

Verified 139 backend tests · 148 frontend tests · 16 live model runs on local Ollama

Date 12 August 2026 · **Hardware** Apple M1 Max, 32 GB unified memory

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What this document is

A record of one work session, in the order the work happened, written so that any of the four of us can present it without having been in the session. Every number in it was measured on this machine and is traceable to a file in the branch; where a number is confounded, the confound is stated next to it rather than left for the audience to find. The speaker script is at the back.

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The gap this closes

The frontend already had a chart contract: seventeen geometries, a long-format frame, and a validator (`validateSpec`) that refuses to draw a figure whose specification the data cannot support. What it did not have was anything that *decides*. A `<Chart>` takes one payload and is deliberately incurious about where that payload came from; the preview page builds its array by hand. So for the two registry datasets the figures were chosen by us, in advance, and for anything else there were no figures at all.

That is the gap. A visitor with their own CSV could not see a single chart, because nothing in the stack could look at an unfamiliar table and work out what it could carry.

THE THEORY WE STARTED FROM, AND WHERE IT WAS WRONG

The original framing was "a new moderator endpoint chooses the graph types". Four parts of that needed correcting before any code was written. There is **no MCP client** in this stack — the one mention of MCP in the repo is a flag that *disables* it. **Chart selection is not the moderator's job**: the moderator rewrites emotional tone and is the one novel thing the project measures, and hanging an unrelated feature off it would make every future tone number measure two changes at once. **"The best three" was not in the contract** and had no landing surface in the interface. And **"any custom dataset" was blocked** on a problem nobody had solved yet: knowing what an arbitrary column *means*.

What survived is the useful half, and it is what got built: a real endpoint, three figures, rendered through the contract the frontend already had. The moderator *model* runs it — that was a later decision, and it is a choice of model, not a merging of stages.

What we built

A Python package of eight modules beside the existing pipeline, and one endpoint. The design principle is a split: what a table *can* carry is a closed question and is computed; which of those figures is *worth showing* is an open question and is asked of a model. The model never names a chart type. It is handed a numbered list of figures that are already drawable, already encoded against real columns and already carrying whatever slice they need, and it picks indices.

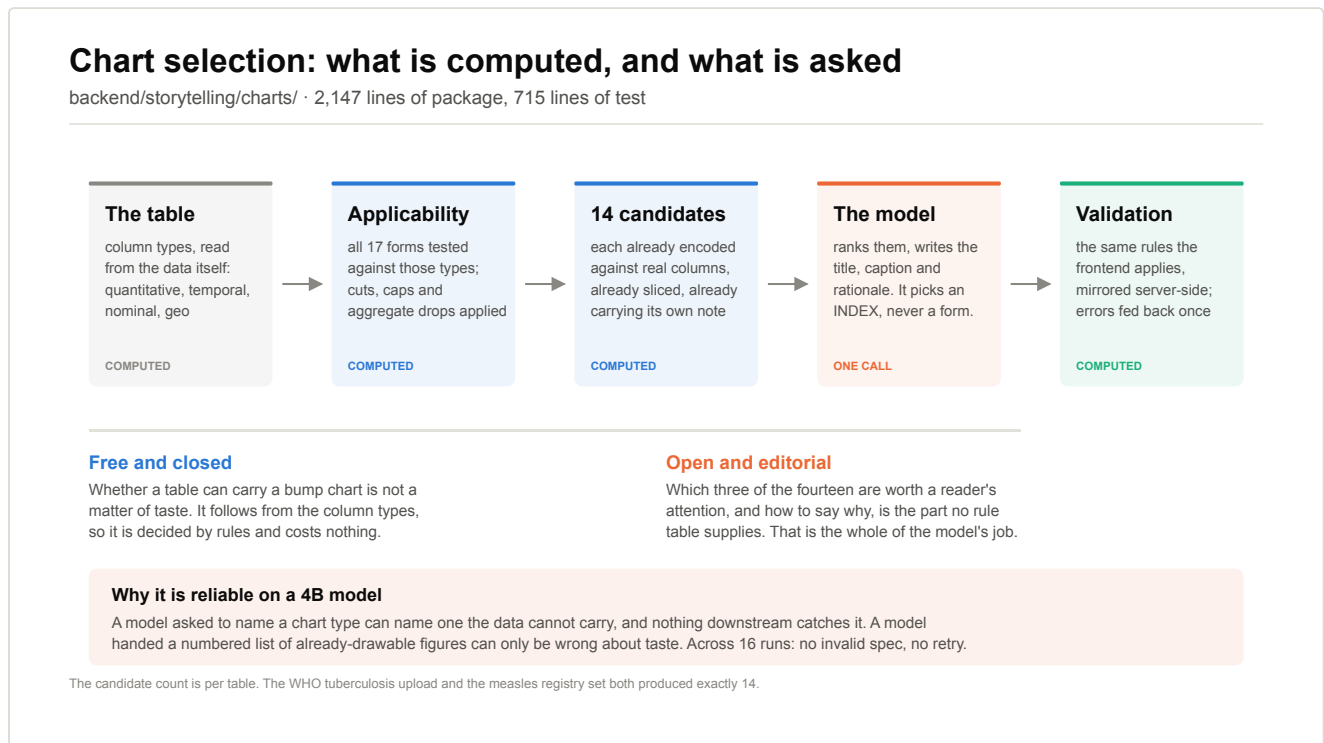


Figure S1. The split the whole design rests on: applicability is computed, ranking is asked.

MODULE	LINES	WHAT IT DOES
spec.py	316	Pydantic mirror of lib/charts/spec.ts. Pinned against the emitted JSON schema by a test, so a contract change on either side breaks the build rather than the page.
validate.py	268	Mirrors validateSpec rule for rule, so a rejected figure can be handed back to the model with the real reason instead of being dropped.
profile.py	362	Types an arbitrary CSV into quantitative / temporal / nominal / geo. The piece with no frontend twin, and the reason an upload works at all.
applicability.py	691	Which of the seventeen forms this frame can carry, and what each needs cut to be honest. The largest module, and the one that does the actual thinking.
select.py	267	The agent: one Ollama call with a schema-constrained response, one retry that feeds the validator's own errors back.
frames.py, http.py	225	Registry datasets and uploads both arrive as a list of frame sources, so there is one selection path rather than two that drift.
tests_charts.py	715	69 tests. Schema parity, validator behaviour, column typing, applicability, the real upload shape, registry-vs-CSV parity, the endpoint.

WHY THIS IS TESTABLE AT ALL

The project's own measurement is that a seeded run reproduces only while the model stays warm, so "the agent picks the same three charts twice" cannot be a test. What is pinned instead is the deterministic layer underneath: column typing, the candidate set, the validation, and the fact that a registry dataset and the same data as an uploaded CSV produce the same candidates. The model's taste is measured by experiment, not asserted by a test.

Three claims we had to walk back

Building the endpoint meant reading what the site already said, and three statements did not survive that reading. None of them were caught by a test, because none of them were code defects — they were claims stronger than the evidence under them.

The causal 0%, reported as a capability wall

The results page carried a red banner: not one of 21 causal claims was supported by the table, therefore causal reasoning is a capability wall rather than a size problem. The first half is a measurement. The second half does not follow from it.

The 0% causal row measures groundedness, not causal ability

Corrected in results.py, app/results/page.tsx and lib/data/metrics.ts

What the page used to say

"Not one of the 21 causal claims across both runs was supported by the table. Causal reasoning is a capability wall, not a size problem."

Printed in a red banner, above every other metric.

What the number can support

Both models state off-table causes with exactly the confidence they state on-table facts.

That is a groundedness finding, and it is the reason the factual check is its own agent.

Why the stronger claim does not follow

The source table is same-day OHLCV

Causation is not one of them

So the row cannot exceed zero

open, high, low, close, volume. Five columns.

"fell amid inflation concerns" points outside the table by construction.

however capable the model is - and a row that cannot exceed zero cannot distinguish a capability ceiling from its own floor.

And the denominator is 21 claims

13 from qwen3.5:4b, 8 from gemma4:12b. By the rule of three, none-correct-out-of-21 is consistent with a true rate near 14%.

Four written records still carry the older wording and were deliberately left alone: RESULTS.md:143, reproductions/REPRODUCTIONS_SUMMARY.md:114, reproductions/paper9-datatales/REPORT_gemma4.md:30, presentation/figures/README.md:41.

Figure S6. The claim we corrected, and the reasoning that replaced it.

WHAT REPLACED IT

The banner is gone. Every metric on the page now carries two lines instead — **what it measures** and **what it cannot tell you** — which is what the original request asked for, and which puts the caveat next to the number it qualifies rather than in a box above all of them. The page also computes the ceiling from the data it has rather than asserting zero: with 21 attempts and none confirmed, an honest upper bound is about 14%.

The home page repeated the same over-claim

The headline statistic said the causal result was "a capability wall for both model sizes". It now states the fact and stops: off-table causes, stated as fact, across the claims attempted, none the table could confirm.

The contract prescribed a bug

Section 7 of `CHART_CONTRACT.md` told the backend to give every modifier an eager default — `stack: "none", orientation: "vertical"`. But `validateSpec` rejects a modifier the form does not honour, and it tests for *presence*, not value. Following the contract as written makes every line chart render as a refusal.

PROVED BEFORE IT WAS CHANGED

Rather than argue from reading, we built the payload the contract prescribed and ran the frontend's own validator over it. It returned `{ok: false, errors: ['line does not honour "stack"', 'line does not honour "orientation"']}`. Modifiers are now `None` server-side and dropped from the wire, and the section was rewritten so the next person to implement against it does not reintroduce the same failure.

Bringing your own table

The dataset picker had a card saying custom uploads were future work. Half of that was still true, and the half that was not is the reason the card became a control: a CSV can be **charted** now, because choosing a form needs only each column's type and a table answers that about itself. It cannot be **generated from**, because a story still needs declared measures and class breaks that no table states.

So the upload response splits those two facts. `chartable` is true, `wired` is false, and the success state offers exactly one thing: see the figures. An upload button that dropped a visitor into a wizard which then refused them would be worse than the placeholder it replaced.

Tested on a table nobody here had seen

The registry datasets are ours; they cannot show that inference works, because their types are declared. So we fetched a third-party medical time series and ran the whole path on it.

PROPERTY	VALUE
Source	WHO tuberculosis incidence and case-detection rate, via Our World in Data
Shape	227 entities · 2000–2024 · 5,597 rows · 2 measures
Missingness	~11% on the second measure
Column types	All inferred. Nothing declared.

FIVE DEFECTS A REAL THIRD-PARTY TABLE EXPOSED THAT OUR OWN NEVER COULD

225 bars on one axis. The cap that exists governs colour, not the category axis, so nothing limited a bar chart's x. Now capped at 15, and the latest year is taken *before* ranking — rank first and it ranks on the wrong number, quite convincingly.

Aggregates ranked as peers. Our World in Data ships World, continents and income groups in the same column as countries. Rows whose code is not ISO3-shaped are now dropped from any figure that ranks places, and the drop is disclosed. This was the one place our own reasoning was wrong first: the note originally said an aggregate outranks everything inside it, which is false for a rate — World sits 59th of 225 on TB incidence per 100k. The real reason is peerhood, not magnitude.

A 225-value colour legend. The scatter bound colour to the country column. 225 hues is not a legend; colour is now omitted when the split exceeds what the form can carry.

A sparse column typed as text. The case-detection rate is absent for a third of country-years and failed a numeric-share threshold, which removed every figure needing a second measure. Types are now read from the non-missing values.

Silent cuts. Only one kind of cut was being disclosed. A bivariate map was quietly showing one year with an empty caption. Every editorial cut now writes itself into the caption the reader sees.

Which model should choose the figures?

By design decision the selector runs on the tier's **moderator** — the agentic role that also runs the factual check. That is a choice of model only: selection remains its own endpoint, and the moderation stage was not touched, so a tone measurement still measures one change.

Which leaves the question of which moderator. Four models were run on the same tables, with the same seed and the same fourteen candidates. The small pair (qwen3.5:4b, gemma4:12b) was measured first; the large pair is below.

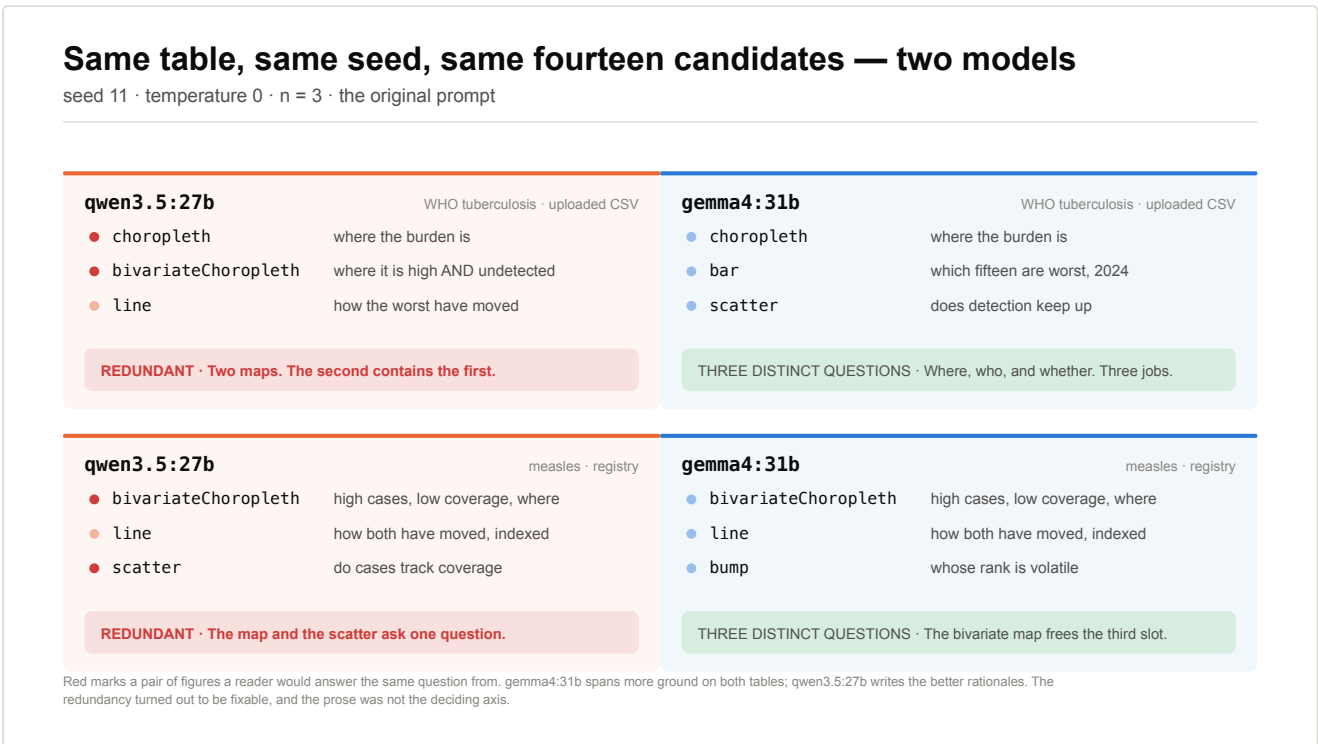


Figure S2. Experiment 1 — two models, same seed, same fourteen candidates, original prompt.

	QWEN3.5:27B · 17 GB	GEMMA4:31B · 19 GB
Question coverage	2 of 4 runs redundant	4 of 4 runs distinct
Rationale quality	Better. Argues form against alternative: "a scatter isolates the statistical relationship better than a map"	Flatter. States the reason without the comparison
Factual accuracy of copy	Two defects (next section)	None found
Compute per selection	44.2 s / 43.7 s	49.0 s / 40.7 s

THE VERDICT, AND WHY IT CHANGED LATER

On the original prompt, **gemma4:31b wins**: it spans more ground and never over-claims. But the deciding axis turned out to be fixable — the redundancy was a prompt defect, not a property of the model, and Experiment 5 closes it for both. What does *not* reverse is the copy behaviour, which is why the recommendation still stands. No tier change is needed: `mid` and `large` already run `gemma4:31b` as moderator, so they inherit this. `qwen3.5:27b` is in no tier and is absent from `experiments/MODELS.md` — it appeared on disk during this session and the roster should record it.

Copy quality, checked against the data

The selector never sees a data value. It is given column names, types, ranges, row counts and the candidate list. So every claim a title makes about what the data *shows* is a claim the model cannot have checked — and we could, by computing it.

Where the smaller-sounding model actually loses: copy, not choice

Three titles from one run, each checked against the table the reader would see

The selector is shown column names, types, ranges and row counts.

It is never shown a single data value. Every claim in a title is therefore a claim it cannot have checked.

"High measles cases often coincide with low vaccination coverage"

Supported, and hedged

$r = -0.310$ across 178 countries in 2023. Weak-to-moderate, and "often" is the right strength of word.

qwen3.5:27b · measles · baseline prompt

"Global measles trends diverge sharply from vaccination rates"

True, by luck

The indexed figure does diverge: cases fall to 18, coverage rises to 235, both indexed to 100 at 1980.

qwen3.5:27b · measles · baseline prompt

"Countries with lower vaccination rates report more measles"

Overstated

Same $r = -0.310$, about 10% of the variance, written as a flat statement of fact with no hedge at all.

qwen3.5:27b · measles · baseline prompt

gemma4:31b, same table, same seed: "Measles cases vs vaccination coverage" · "Global trends in measles measures"

Duller, and it asserts nothing it has not been given. On a project measuring over-claiming, that is the cheaper failure.

Correlations computed from the project's own measles frame at 2023, the year those figures slice to. The 2024 slice gives $r = -0.165$ with $n = 180$; the sign is stable, the strength is not.

Figure S4. Experiment 2 — three titles checked against the table the reader would see.

The second defect is smaller but sharper. qwen3.5:27b titled the all-years tuberculosis map "*...by country in 2024*". That is not false — the scrubber opens on the last year — but 2024 appears nowhere in that candidate's listing. It was borrowed from a sibling candidate the model also picked, against an explicit instruction to state no number that was not given, and it undersells a figure carrying 25 years and a play button. gemma4:31b, same seed, titled it without a year and wrote "over time" in the rationale.

WHY THIS IS THE FAILURE MODE THAT MATTERS HERE

This project measures whether a moderator can lower a story's alarmism without flattening real urgency. A model that writes conclusions it has not verified is producing exactly the thing the project exists to measure — one layer up, in the furniture around the story rather than the story itself. Duller copy is the cheaper mistake.

Throughput, and a contaminated clock

The first set of timings was nonsense and it took a process listing to notice: another session was running `experiments/run_repeats.py` against the same Ollama server. The same selection measured 372.0 s, then 143.2 s, then 63.2 s once the competing job died — with byte-identical output all three times.

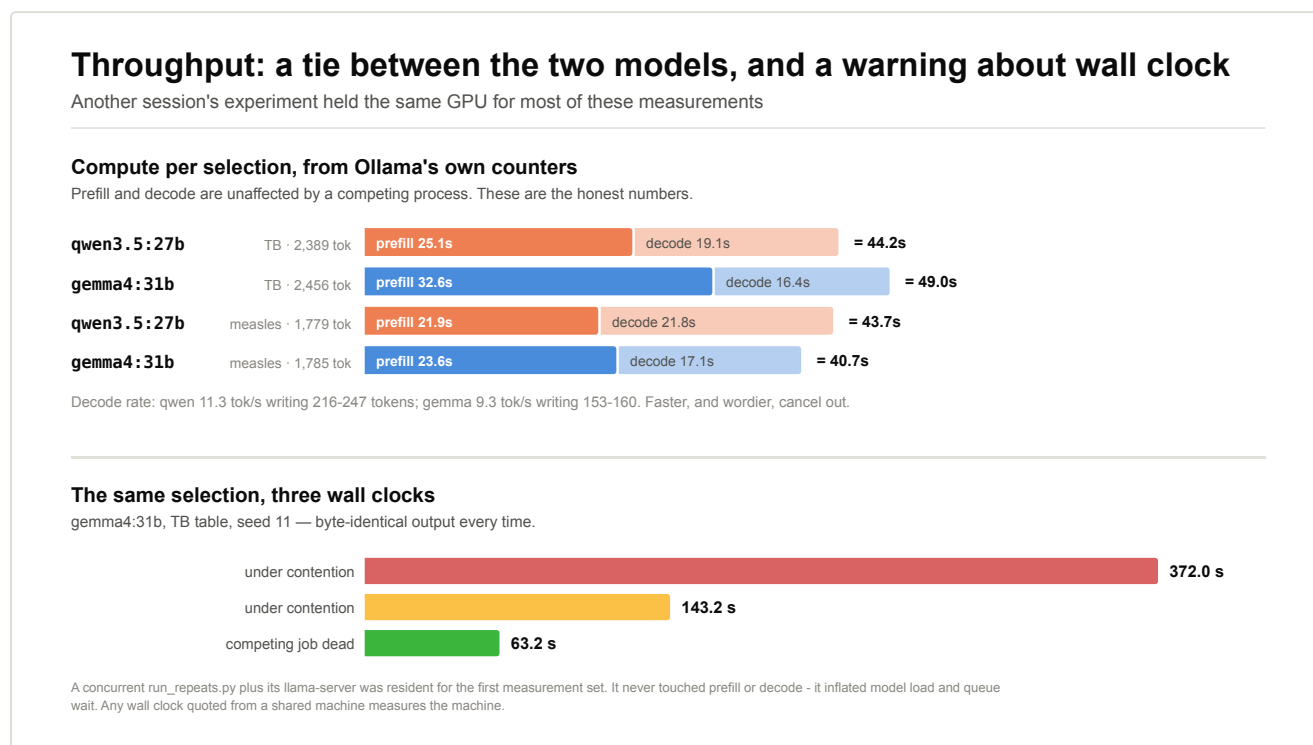


Figure S3. Experiment 3 — compute is a tie; the wall clock was measuring another session's job.

THE METHOD WORTH REPEATING

Ollama returns its own counters on every response: `load_duration`, `prompt_eval_duration`, `eval_duration`. Contention inflates load and queue wait; it leaves prefill and decode alone. Measured against a clean run afterwards, prefill moved by 0.6 s and decode by 0.1 s — so the counters were telling the truth while the wall clock was off by a factor of six. **Any wall clock quoted from a shared machine is measuring the machine.**

On the honest numbers the two models are a tie: `qwen3.5:27b` decodes about 20% faster and writes about 40% more, and the two cancel. What actually separates them in practice is model load, which depends on cache state and on who else is using the GPU.

A seed that survived an eviction

An accidental result, and it qualifies a finding already written into RESULTS.md. That finding is that seeded reproducibility holds inside a warm process and fails across a cold load, which is why the committed raw stories are treated as the only reproducible artefact.

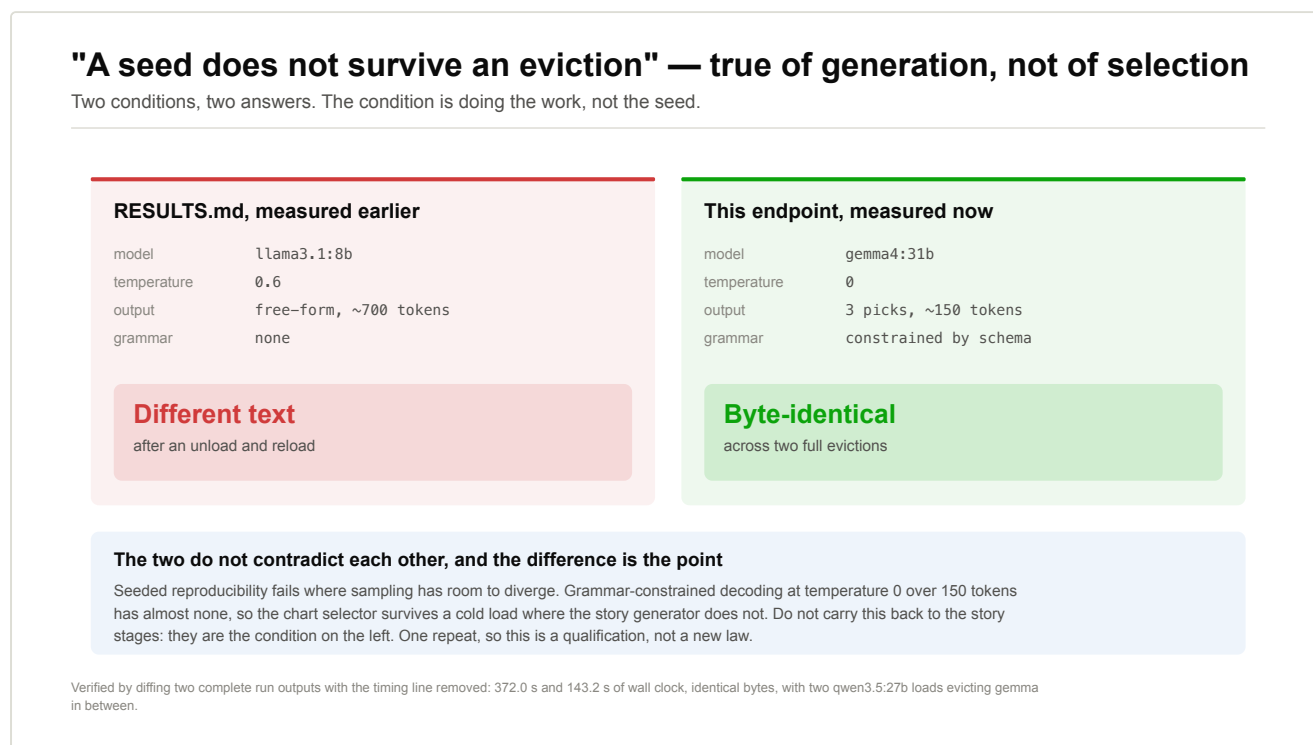


Figure S5. Experiment 4 — the standing reproducibility finding, and the condition where it does not apply.

It still holds for generation. It does not hold here. Two complete runs of the same selection were diffed in full — identical bytes, with two 17 GB model loads evicting the model in between. The conditions are different in three ways at once, and any of them could be doing the work: temperature 0 rather than 0.6, grammar-constrained decoding rather than free text, and about 150 output tokens rather than 700.

DO NOT OVER-READ THIS

One repeat, one model, one table. It is a qualification of a stated finding, not a replacement for it, and it must not be carried back to the story stages — those are the left-hand condition in the figure. If it is worth a line in the paper, it needs the three variables separated first.

Redundancy was a prompt defect — and prose could not fix it

Experiment 1 left one clear defect: on two of four runs a model spent two of its three slots on figures a reader would answer the same question from. Two maps of the same measure; a bivariate map and a scatter of the same two measures. The hypothesis was that the instruction was at fault. It said "prefer variety", and variety of *form* is not what was wanted.

Redundancy was a prompt defect, and prose was not enough to fix it				
Twelve runs · two models · two tables · three prompts · seed 11 throughout				
	qwen · TB	qwen · measles	gemma · TB	gemma · measles
Baseline "Prefer variety: three near-identical figures waste the reader's time." 2 of 4 clean	✗	✗	✓	✓
Variant A · prose "Each figure must answer a DIFFERENT question... judge that by the question, not the shape." 2 of 4 clean	✗	✗	✓	✓
Variant B · schema A required question field, decoded BEFORE the index, plus the same instruction. 4 of 4 clean	✓	✓	✓	✓
Why prose could not do it The model had no place to put a question, so it had nothing to compare the next figure against.		And a second thing fixed itself Every over-claiming title disappeared. Asked for the question first, both models titled the question, not an answer.		

Figure S7. Experiment 5 — three prompts, twelve runs. The schema change is what worked.

Variant A — say it properly

The instruction was rewritten to be about the question a reader walks away having answered, explicitly not about the shape of the figure. It changed almost nothing. qwen3.5:27b returned the same three forms on tuberculosis in a different order, and on measles swapped a scatter for a connected scatter — the same two measures against each other, with time added. Still 2 of 4 clean.

Variant B — give it somewhere to put the answer

The second attempt changed the response schema rather than the wording. Each pick now carries a required question field, declared **before** the index. Fields are decoded in schema order, so the model must state a question and then find a figure that answers it, instead of taking a figure and reverse-engineering a justification — and, crucially, it can see its own earlier questions while writing the next one. That is the thing prose could not provide: a place to compare against.

4 OF 4 CLEAN, AND TWO RESULTS NOBODY WAS AIMING FOR

Both models converged on the same three questions. On both tables, independently: how has this moved over time, which places are worst now, and do the two measures relate. For a country × year × two-measure table that is arguably the right trio, and two different model families finding it from a list of fourteen is a stronger signal than either finding it alone.

Every over-claiming title disappeared. Not asked for, not expected. Asked to write the question first, both models titled the question rather than an answer: "Vaccination Coverage vs. Measles Cases in 2023" instead of "Countries with lower vaccination rates report more measles". The defect in Experiment 2 turns out to have been partly structural — a model with nowhere to put the question puts it in the title, as an assertion.

ONE REGRESSION, UNFIXED

Nine of the twelve variant-B titles came back in Title Case, against the sentence case used everywhere else in the interface. Cosmetic, one line of prompt, and not worth another four minutes of model time before the talk — but it should be fixed before this ships.

RUN	BASELINE	VARIANT A · PROSE	VARIANT B · SCHEMA
qwen · TB	choropleth, bivariateChoropleth , line	choropleth, line, bivariateChoropleth	line, bar, bivariateChoropleth
qwen · measles	bivariateChoropleth , line, scatter	bivariateChoropleth , connectedScatter , line	line, bar, scatter
gemma · TB	choropleth, bar, scatter	choropleth, bar, scatter	bar, choropleth, scatter
gemma · measles	bivariateChoropleth, line, bump	choropleth, scatter, line	bivariateChoropleth, lollipop, line

Bold marks the members of a redundant pair. All sixteen runs across all three prompts: one model call each, no grammar fallback, no validator rejection, no retry.

What is still open

1. **No pull request yet.** The branch carries a teammate's 14 chart-contract commits plus 8 of ours. If feat/chart-contract merges first the PR reads as 8 focused commits; if not, reviewers see all 22 and re-review work that was already reviewed.
2. **Four written records still say "capability wall."** Deliberately left alone, because they are the research record and changing a stated finding there is not a silent edit to make: RESULTS.md:143, reproductions/REPRODUCTIONS_SUMMARY.md:114, reproductions/paper9-datatales/REPORT_gemma4.md:30, presentation/figures/README.md:41. Note the last one is the speaker script for the existing fig2 — **whoever presents that figure will say the corrected claim's older form out loud** unless it is fixed first.
3. **Title Case regression** from variant B, above.
4. **The uploaded file still cannot be generated from.** Charting an arbitrary table needs column types; generating a story from one needs declared measures and class breaks. That gap is real and is stated in the interface rather than hidden.
5. **The tone axis is still single-axis** (alarmism only). Unrelated to this work and not started.

Speaker script

Written to slot beside the existing eight-figure Results section, which runs about 7 minutes of content plus a 3.5-minute demo. This material adds roughly **4 minutes** and one demo beat. If the talk is already full, sections 9 and 3 are the two that earn their place; the rest is backup for questions.

BEAT	FIGURE	TIME	POINT TO LAND
A	figS1	50 s	The model never names a chart type
B	figS7	70 s	A schema change fixed what a prompt change could not
C	figS6	55 s	We corrected one of our own published claims
D	figS4	45 s	The model over-claims where it cannot check
—	demo	60 s	Upload a CSV nobody has seen, get three figures
Q&A	figS2, figS3, figS5	—	Held in reserve

Beat A — figS1 · The division of labour

"Seventeen chart types. The question of which ones a table *can* carry is not a matter of taste — it follows from the column types, so we compute it. What the model gets is a numbered list of figures that are already drawable, already bound to real columns, already carrying whatever cut they need. It picks three indices and writes the copy. It never names a chart type. That is why this runs on a 4-billion parameter model without producing a single invalid figure across sixteen runs."

If asked "why not just let the model choose freely": because a model that names a form can name one the data cannot carry, and nothing downstream catches it. Here the worst it can be is wrong about taste.

Beat B — figS7 · The variety experiment

"The first version had a real defect: on half the runs, two of the three figures answered the same question. Two maps of the same measure. So we rewrote the instruction to say exactly that — judge sameness by the question, not the shape. It changed nothing. Same forms, different order. What worked was not asking better: it was adding a required *question* field to the response, decoded before the figure index. Now the model has to state a question, and it can see the questions it already wrote. Four out of four clean. And two things we were not aiming for: both models converged on the same three questions — how has this moved, who is worst now, do the two measures relate — and every over-claiming title disappeared, because a model asked for the question first stops putting the answer in the headline."

Caveat to state: twelve runs, two models, two tables. That is an effect worth acting on, not a benchmark.

Beat C — figS6 · The causal 0%

"Our own results page said the zero-percent causal score proved a capability wall. It does not, and the reason is structural. The source table is same-day open, high, low, close, volume. Causation is not a column in it. So a claim like 'fell amid inflation concerns' points outside the table and cannot be supported there however capable the model is — which means the row cannot exceed zero, and a row that cannot exceed zero cannot

tell a capability ceiling from its own floor. What the number does show survives, and it is the finding that matters: both models state off-table causes with exactly the confidence they state on-table facts. That is groundedness, and it is why the fact-checker is a separate agent."

Caveat to state: and the denominator is 21 claims. By the rule of three, none-out-of-21 is consistent with a true rate near 14%.

Beat D — figS4 · Titles the model cannot check

"The selector is shown column names, types and ranges. It is never shown a single value. So we took three titles it wrote and computed whether they were true. One was supported and correctly hedged. One was true by luck. One asserted a correlation of minus zero-point-three-one — about a tenth of the variance — as a flat statement of fact. That is the same failure the whole project is about, one layer up: confidence that is not earned by the evidence underneath it."

Do not say the model "hallucinated". It made claims it had no way to check, which is a different and more interesting failure.

Demo beat

"This is a WHO tuberculosis table — 227 countries, 25 years, two measures, and nobody in this project has ever configured it. Upload. The column types are read from the data. Three figures. And read the small print under them: it tells you the top fifteen were kept, that the World row was dropped because an aggregate is not a peer of the countries inside it, and which year is shown. Every editorial cut says so."

Have ready: the file is at `.playwright-mcp/who_tuberculosis_owid.csv`, and the backend needs `DJANGO_CORS_ORIGINS` set for the browser upload.

Questions we should expect

QUESTION	ANSWER
Is this the moderator?	It runs on the moderator <i>model</i> , by design decision. It is a separate stage and separate endpoint, so a tone measurement still measures one change.
Which model is better?	<code>gemma4:31b</code> , on copy accuracy. On figure choice the gap closed once the schema was fixed. <code>figS2</code> .
How fast is it?	About 45 seconds of compute on a 30B model, plus load. Do not quote wall clock from a shared machine. <code>figS3</code> .
Is it reproducible?	Byte-identical across an eviction here, which is <i>not</i> true of the story generator. One repeat. <code>figS5</code> .
Can I upload anything?	Any CSV with at least one numeric column. It will be charted, not turned into a story — and the interface says which.